

Neuronal structure and function

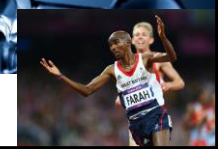
REFERENCES:

See Pocock, Richards & Richards 4th edition, Ch 7 (99-106) & Ch 9 (141-150).

5th edition:Ch 7 (101- & Ch 9 147-)

Koeppen & Stanton 6th edition
Berne & Levy Physiology. Ch 5

Stanfield & Germann 3rd edition. Chapter 7



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- m.mayston@ucl.ac.uk

Neuronal structure and function

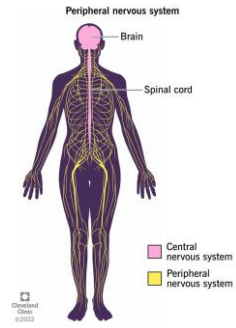
At the end of this session you should be able to :

- **Understand and be able to name the basic anatomical divisions of the nervous system**
- **Name and state the function of the main cells of the nervous system: neurons and glia**
- **Describe the structure of nerve cells (neurons) and their projections (axons/nerves)**
- **Describe the action potential (AP) in terms of:**
  - **its ionic basis**
  - **threshold and refractory period**
  - **Propagation**

Neuronal structure and function









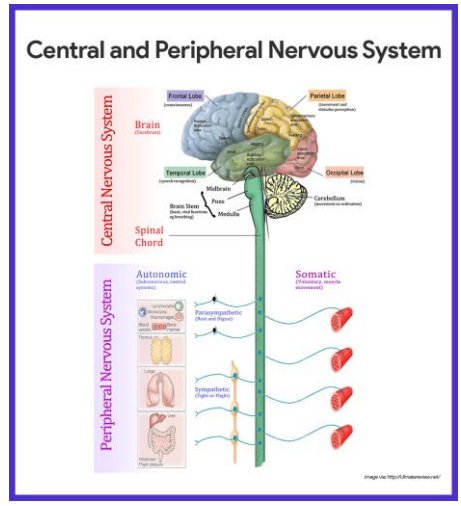




Neuronal structure and function

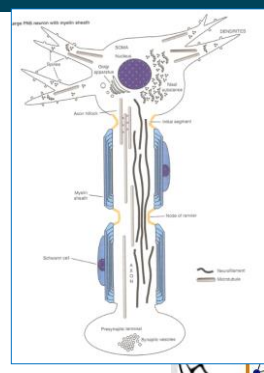
Key points:

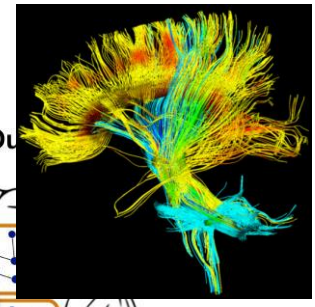
- The basic anatomical organization of the nervous system
- The cells of the nervous system
- The ionic basis of the action potential
- The conduction of the action potential

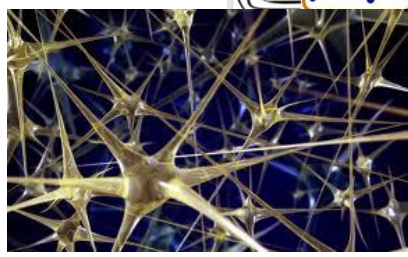


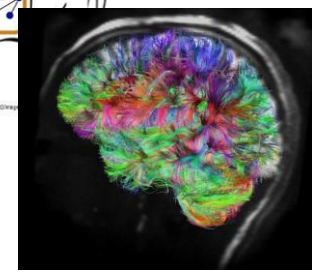
4

Neuronal structure and function



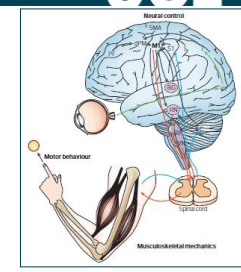






5

1. The basic anatomical organisation of the nervous system



Anatomical divisions of the nervous system

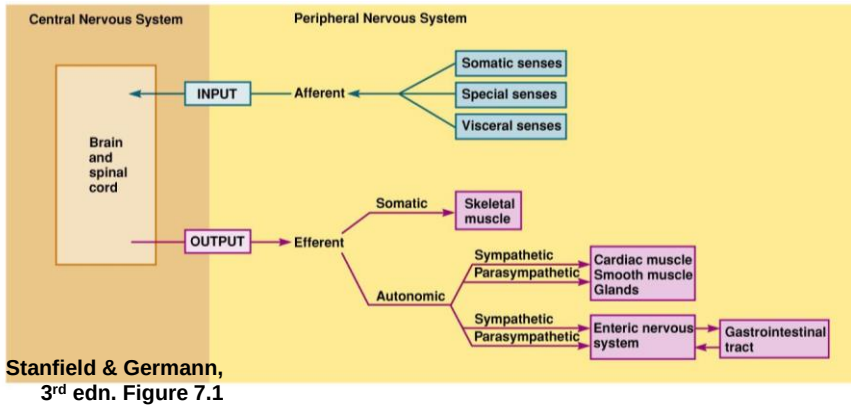
- The brain
- The spinal cord
- The peripheral nerves
- The autonomic nervous system
- The enteric nervous system (controls activity of the gut)



6

1. The basic anatomical organisation of the nervous system

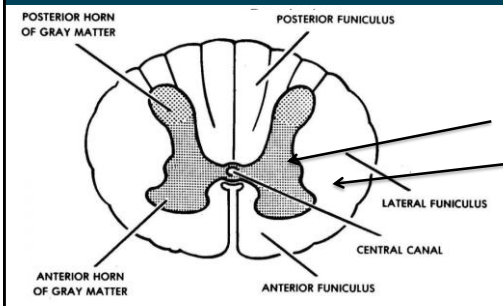
Overview of the Nervous System



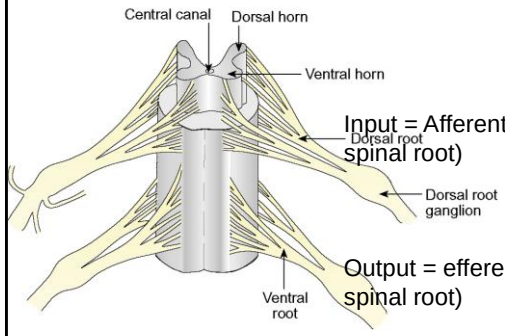
Stanfield & Germann, 3rd edn. Figure 7.1

7

1. The basic anatomical organisation of the nervous system



Grey (cell bodies) and white matter (myelinated axons) in the spinal cord



Input = Afferent (dorsal spinal root)

Output = efferent (ventral spinal root)

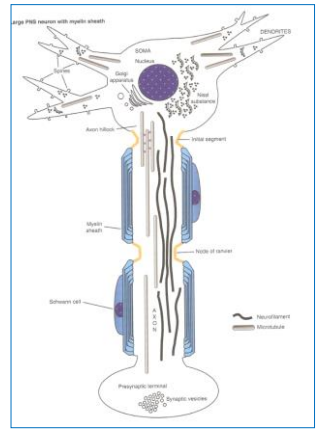
8

2. The cells of the nervous system

UCL

Key points:

- The basic anatomical organisation of the nervous system
- The cells of the nervous system: 1. neurons (nerve cells) and 2. glia (glial cells)
- The ionic basis of the action potential
- The conduction of the action potential



Neurons are the basic unit of the nervous system: motor, sensory and most prolific are the interneurons. They are supported by the glial cells which outnumber all of the neurons.

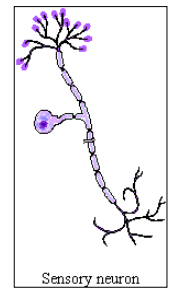
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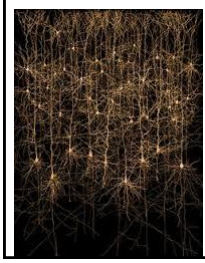
2. The cells of the nervous system

UCL

The CNS is made of two main types of cell:

- Nerve cells or neurons
  - have characteristic thread-like extensions: dendrites at the input end and axon at the output
  - Different types: motor, sensory, interneurones
  - Cell bodies found in grey matter of CNS and in autonomic ganglia
  - different morphological types
  - Are ‘excitable’ cells: muscle cells (myocytes) are also excitable cells.


Sensory neuron



Glial cells or neuroglia

10

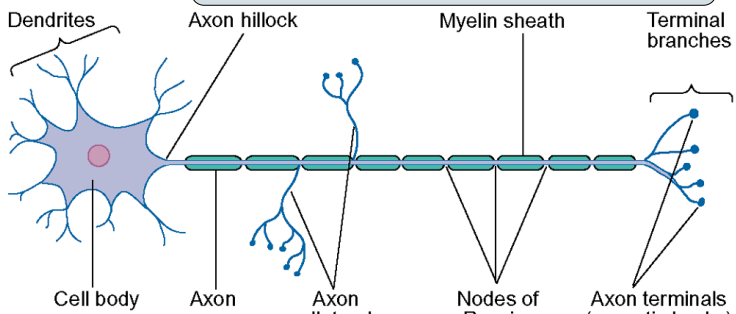
2. The cells of the nervous system

UCL

Main structural features of neurons

Dendrites = input

Integration: APs generated at/around axon hillock; trigger zone



Axons = communicate / output including via collaterals

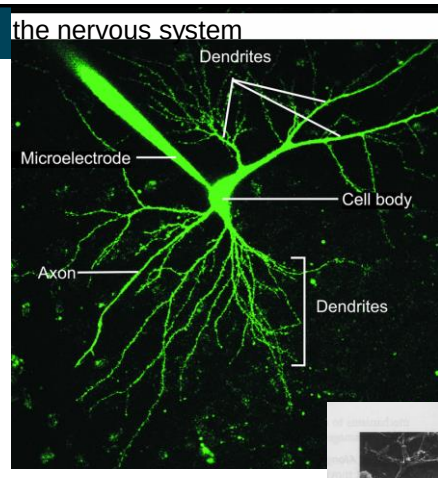
Cell body
'Soma'
Metabolic engine for neuron

11

2. The cells of the nervous system

UCL

Multiple dendrites- but only one axon!



P.R.&R.

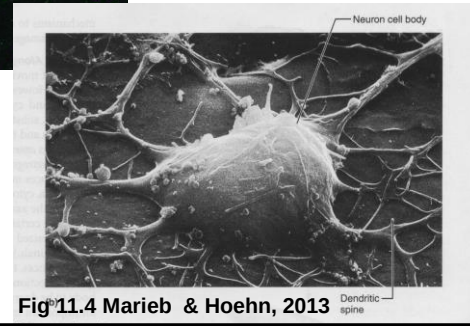
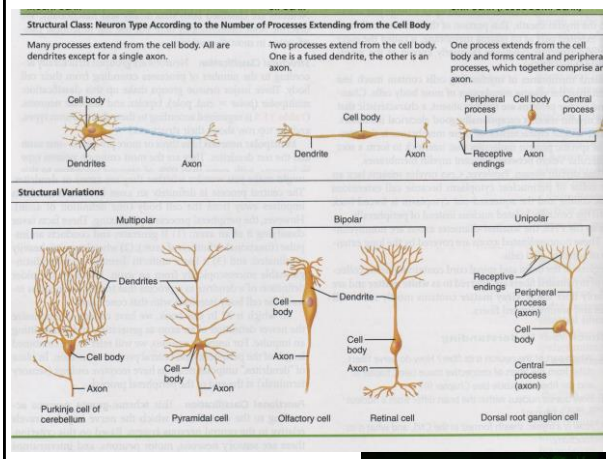
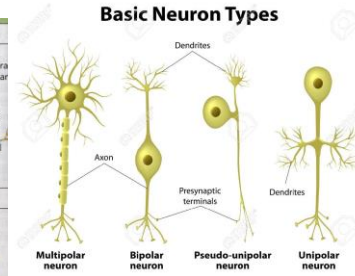
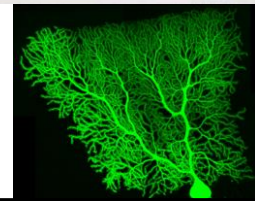


Fig 11.4 Marieb & Hoehn, 2013

2. The cells of the nervous system

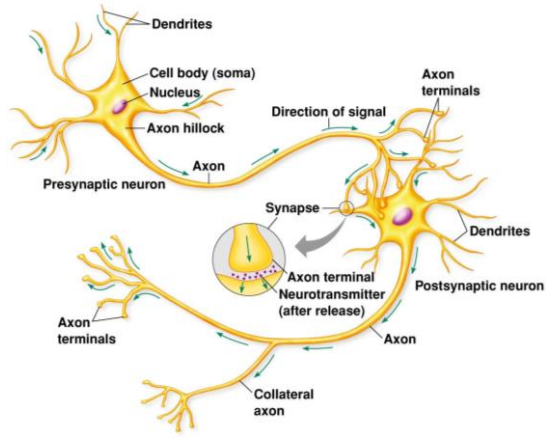






13

Components of a Neuron: transmission between neurons (**next lecture**)



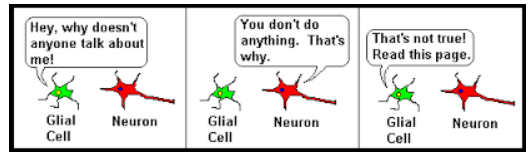
S&G: Figure 7.2

Neuron to neuron: 'power' of a neuron depends on its connections!

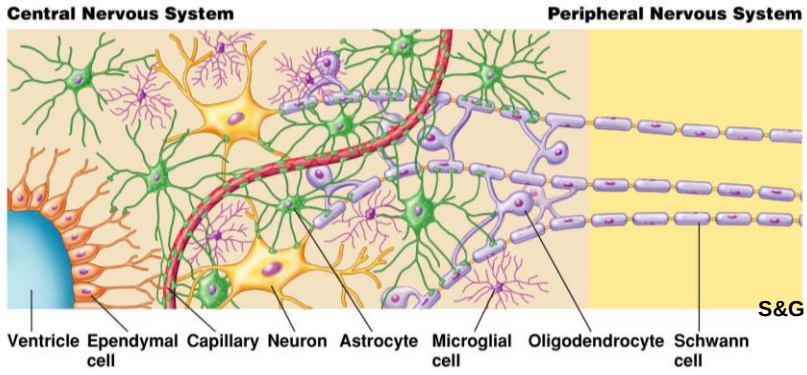
14

The CNS is made of two main types of cell:

- *Nerve cells or neurons*
  - *different morphological types*
  - *have characteristic thread-like extensions: dendrites at the input end and axon at the output*
- Glial cells or neuroglia:
  - The support cells of the CNS, and far outnumber neurons (~90% of all cells in the nervous system)
  - 4 main types: astroglia, oligodendrocytes, microglia, ependymal cells (in the PNS Schwann cells and macrophages). Oligodendrocytes and Schwann cells are crucial to neurotransmission



15



S&G Figure 9.1

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Ependymal cells- line central fluid filled spaces e.g. ventricles of brain, central canal of spinal cord

Astroglia: attach to blood vessels

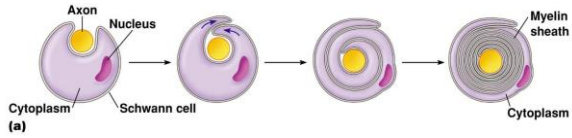
Oligodendrocytes/ Schwann cells

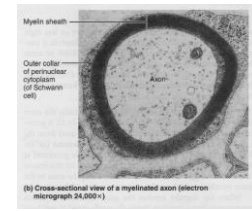
Microglia: phagocytic- converge of site of injury or infection

16

Schwann cell/Oligodendrocyte: Myelin Sheath

Figure 7.5; S&G





CNS

PNS

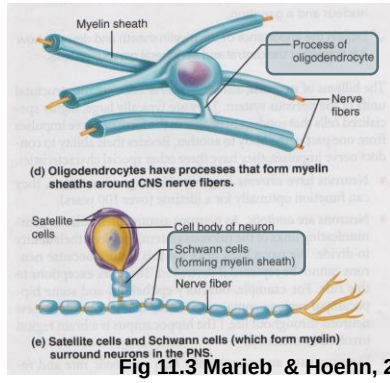
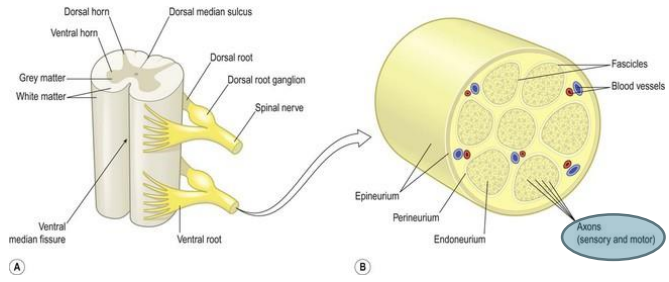




Fig 11.3 Marieb & Hoehn, 2013

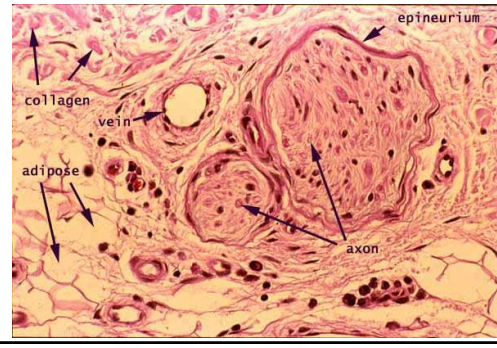
17



Axons in bundles – ***fascicles***- anchored to connective tissues by **epineurium**

Each bundle covered by **perineurium**

Endoneurium: covers individual nerve fibres



18

Neurons: summary 1

- Neurons (nerve cells) are the basic cell of the nervous system and are supported by glial cells which outnumber the neurons
- The main function of the neuron is to transmit information to other neurons (nerve cells) or to the neuromuscular junction via its axon: neurons are ***excitable*** cells (separation of electrical charge across the cell membrane) and capable of producing large rapid electrical signals called action potentials (AP)
- In the PNS axons (if insulated are surrounded by Schwann cells) run in peripheral nerve trunks alongside blood vessels
- Nerves (bundles of axons) can be sensory, motor or mixed
- Peripheral nerves can also contain sympathetic postganglionic fibres which innervate muscles and glands

KEY POINT: "The power of a neuron depends on its connections"

19

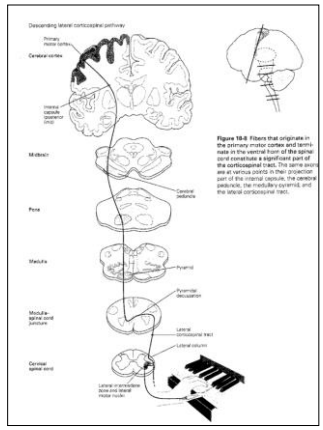
Neuronal and nerve structure and function: Learning outcomes

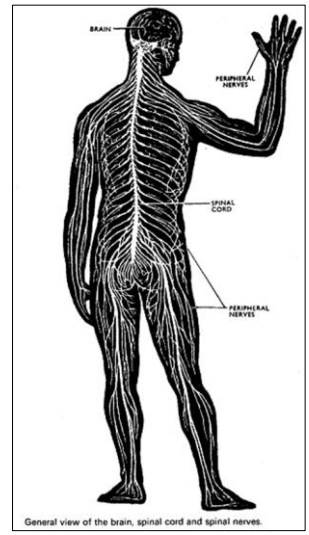
- Be aware of and able to name the basic anatomical divisions of the nervous system
- Describe the structure of nerve cells (neurons) and their projections (axons/nerves)
- **Describe the action potential (AP) in terms of:**
  - **its ionic basis**
  - **threshold and refractory period**
  - **Propagation**



20

Signal transmission to effect action: Why action potentials?

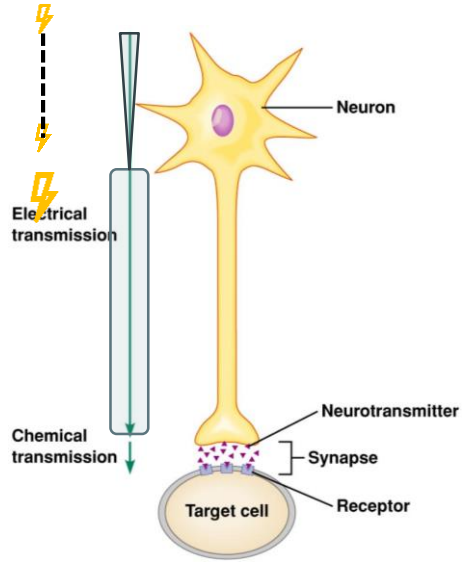




General view of the brain, spinal cord and spinal nerves.

21

Signal transmission in neurons: ***structure = function***



S&G: Figure 5.19

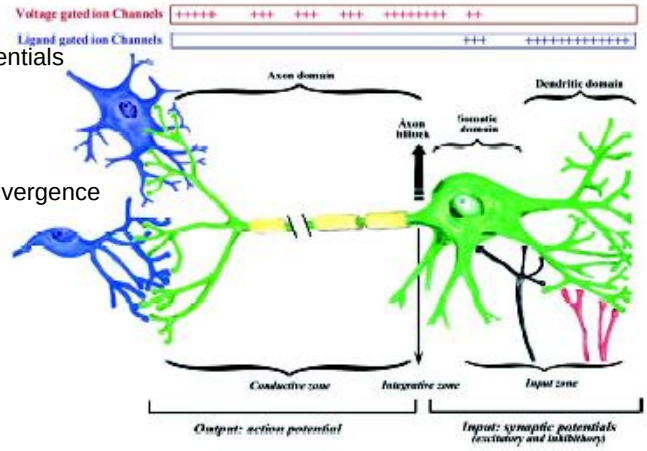
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Signal transmission in neurons: ***structure = function***

Action potentials

Graded potentials

Divergence

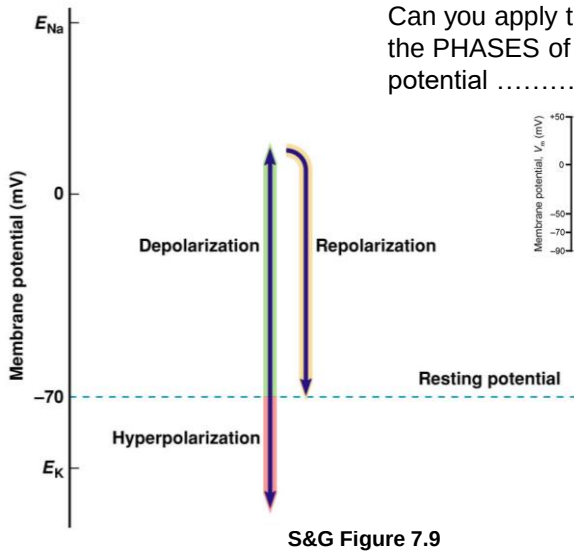


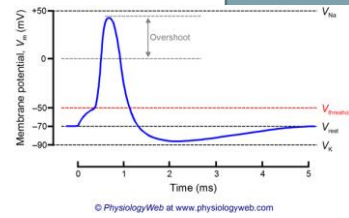
Giuliodori & DiCarlo, 2004

23

Signal Transmission associated with ***Membrane Potential Changes*** along the axon

Can you apply these terms to the PHASES of the action potential ……?????

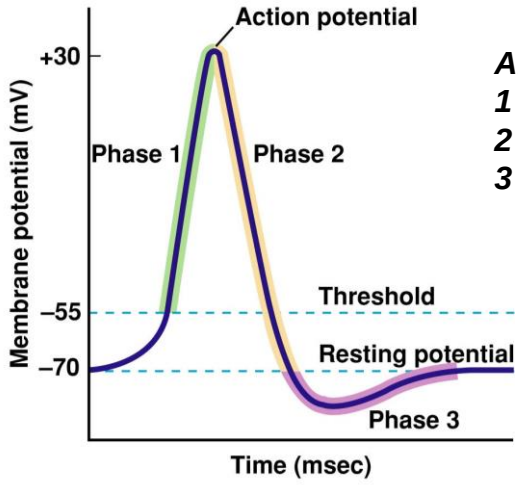




S&G Figure 7.9

24

Phases of an Action Potential



(a) Three phases of an action potential

S&G Figure 7.13a



ANSWER:
1 = Depolarisation
2 = Repolarisation
3 = Hyperpolarisation

25

Key facts: action potential

- Neurons (nerve cells) require a certain minimum depolarization before they will generate an action potential. This is known as the ***threshold.***
- ***Changes in ionic permeability*** underlie the changes in membrane potential associated with the action potential- determine **shape** and **duration** of the AP.
- Each action potential is similar in amplitude and duration: action potentials are described as being ***all or none*** in character. The AP is **self propagating.**
- After an action potential has passed along an axon it cannot conduct another until a certain time has elapsed - axons have a ***refractory period, both absolute and relative.***
- The ***frequency*** of action potentials codes for stimulus intensity.

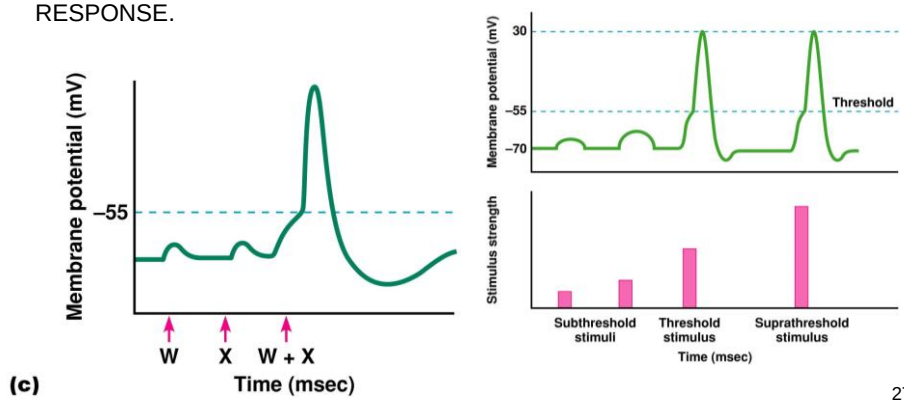
26

The action potential: threshold for activation

Generation: requires an adequate stimulus (e.g. specific sensory stimulus; next lecture) i.e. there is a minimum strength required = threshold to open enough ion channels to bring the membrane potential to threshold

If below = subthreshold = no action potential

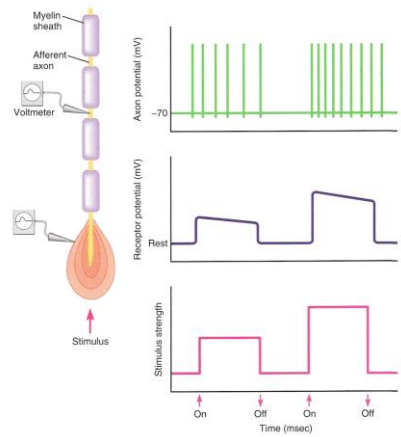
If above = suprathreshold = generates action potential: but, the magnitude of the AP is the same as for a threshold stimulus because it is an ALL OR NONE RESPONSE.



27

Action potential: rate coding determines stimulus intensity

Coding of stimulus intensity: firing frequency depends on the duration of the stimulus and the absolute refractory period of the AP.



Strength of stimulus coded by frequency of APs

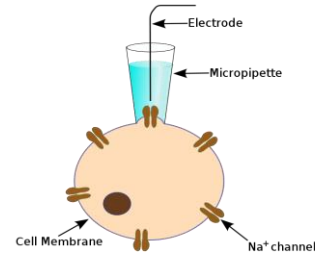
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UCL

Key points:

- The basic anatomical organization of the nervous system
- The structure of nerve cells
- The ionic basis of the action potential
- The conduction of the action potential

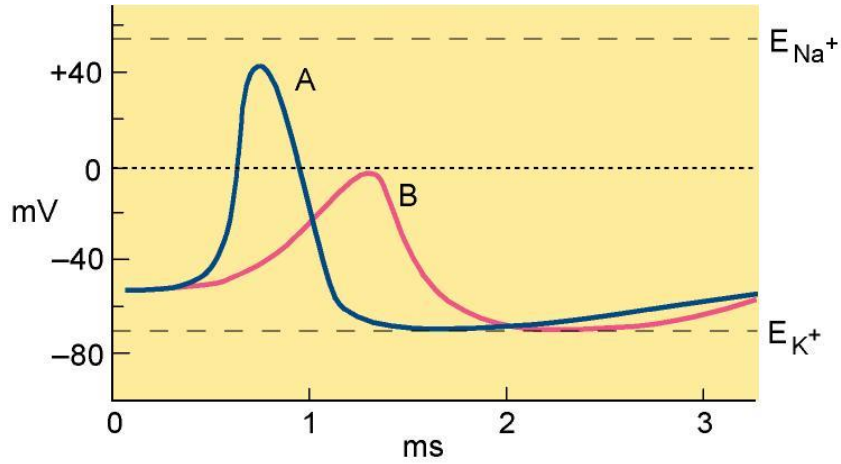
Fast
Unidirectional
Simple
Effective



29

UCL

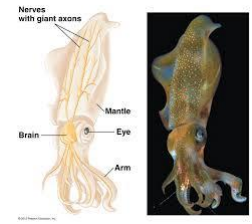
The action potential amplitude is sodium dependent



A - normal Na^+ concentration; B - Na^+ concentration reduced by 66%

30

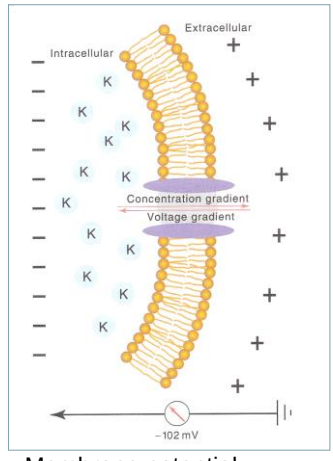
Action potentials



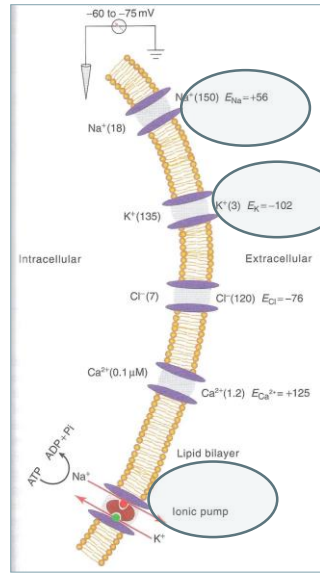
- Basis of APs are **voltage dependent** ion channels
- APs are propagated with **the same size and shape** along an axon
- AP propagation along the **axon** is unidirectional
- APs are usually initiated at the axon hillock (junction of cell body and axon); **all or none**
- APs are the basis of the signal carrying ability of neurons; **self propagating**
- Patterns of APs i.e. ***frequency,*** encode information conveyed by neurons
- The signature feature of a neuron is its ability to **rapidly change its membrane potentia**l in response to an appropriate stimulus – *initiated by changes in the permeability of* Na^+

31

Neuronal action potential



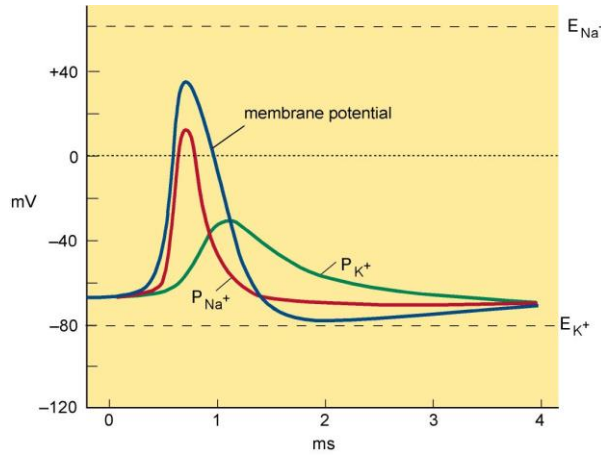
Membrane potential:
1. Chemical = concentration gradient
2. Ion permeability= voltage gradient
1+2=electrochemical gradient



Channel types:
Leak channels
Voltage gated channels
Ligand gated channels

Action potential: Mechanisms of generation:

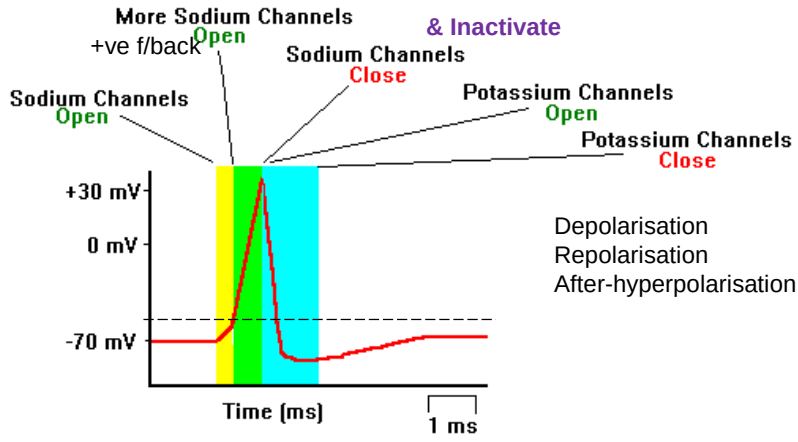
Changes in permeability of Na^+ and K^+ by opening and closing of voltage gated channels determine the phases of the AP



33

Action potential: Mechanisms of generation

Sequence of opening and closing of sodium and potassium channels:



Depolarisation
Repolarisation
After-hyperpolarisation

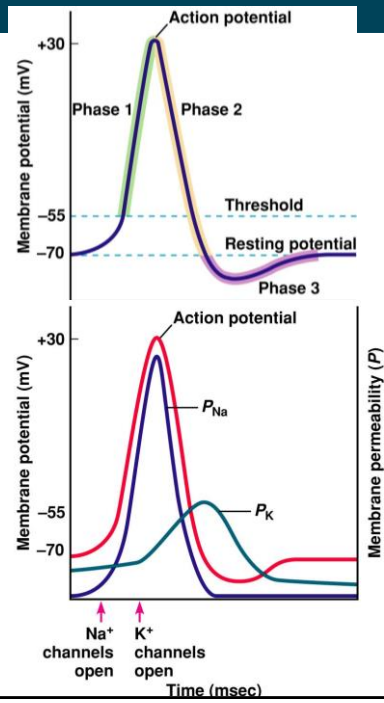
Ion channel activity during the AP

34

Action potential and changes in ionic permeability

Many more leaky K^+ channels than Na^+
Voltage gated (VG) K^+ channels open later than VG Na^+ channels and are open for longer
Unlike Na^+ channels K^+ channels do not have an inactivation gate

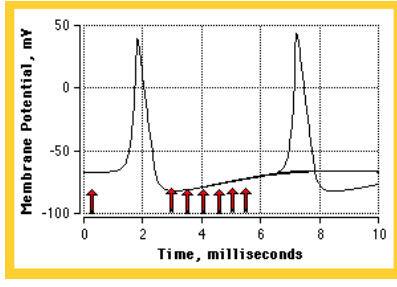
Key point: Timing of opening/closing of voltage gated channels



35

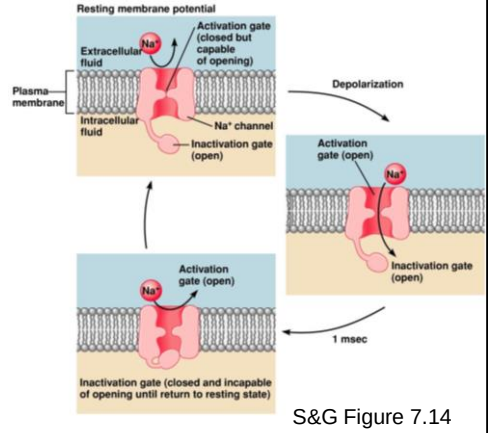
Action potential: refractory period

Refractory periods: K^+ channels slower to open and close than Na^+ channels; Na^+ channels become ***inactive*** after opening: consequences of this?



Absolute Relative

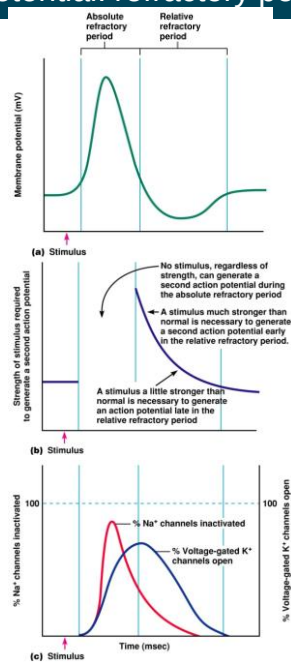
After Hodgkin & Huxley, 1952



S&G Figure 7.14

36

Action potential: refractory period



Refractory periods contribute to:

All or none property

Frequency of firing i.e. coding of information

Unidirectional propagation along **axons** but can back propagate in ***dendrites***! (See Gasparini et al 2007; Häusser et al 2000; Häusser group 2013)

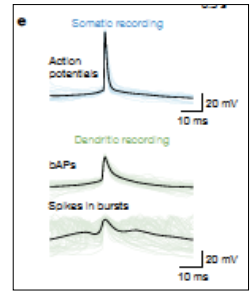
APs can only travel the wrong way along ***axons*** in response to **artificial stimulation** of axons – antidromic transmission.

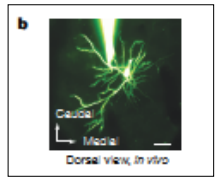
37

UCL research focus!!!!!!!!

Action potential generation and synaptic function. Prof Michael Häusser

"Neuronal dendrites express voltage-dependent Ca^{2+} and Na^{+} channels that confer electrical excitability, particularly the ability to support the active back-propagation of action potentials and the initiation of local dendritic spikes" In Smith et al from Hausser Lab at UCL, Nature, 2013





38

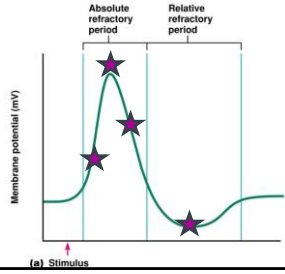
Action potential:

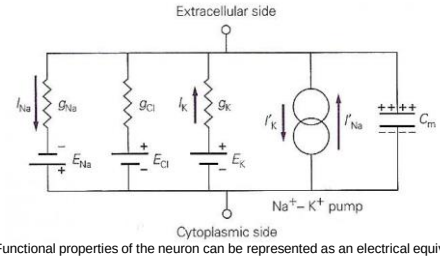
Revision! However, the membrane potential of the neuron is determined by the ionic gradients and permeability of the membrane to **several** different ions

The Goldman Hodgkin Katz (GHK) Equation

$$E = \frac{RT}{F} \ln \frac{P_{Na}[Na^+]_o + P_K[K^+]_o + P_{Cl}[Cl^-]_i}{P_{Na}[Na^+]_i + P_K[K^+]_i + P_{Cl}[Cl^-]_o}$$

P_{Na}, P_K etc. are the permeability constants of the membrane to Na^+, K^+ etc. $[Na^+]_o$, $[Na^+]_i$ etc. are the extracellular and intracellular concentrations of the individual ions





Functional properties of the neuron can be represented as an electrical equivalent circuit.

39

Key facts: action potential

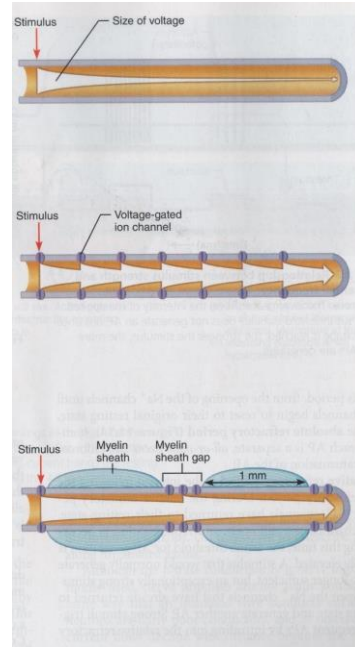
- Neurons (nerve cells) require a certain minimum depolarization before they will generate an action potential. This is known as the ***threshold.***
- ***Changes in ionic permeability*** underlie the changes in membrane potential associated with the action potential.
- Each action potential is similar in amplitude and duration: action potentials are described as being ***all or none*** in character. AP is **self propagating.**
- After an action potential has passed along an axon it cannot conduct another until a certain time has elapsed - axons have a ***refractory period.*** Also results in unidirectional propagation (NB dendritic APs).
- The ***frequency*** of action potentials codes for stimulus intensity.

40

Action potential:

Key points:

- The basic anatomical organisation of the nervous system
- The structure of nerve cells
- The ionic basis of the action potential
- The conduction of the action potential



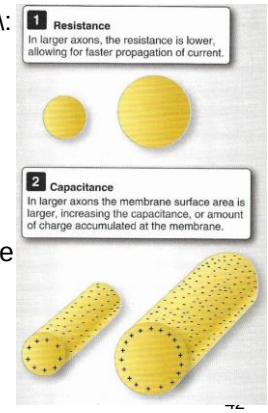
Action potential: Mechanisms of conduction

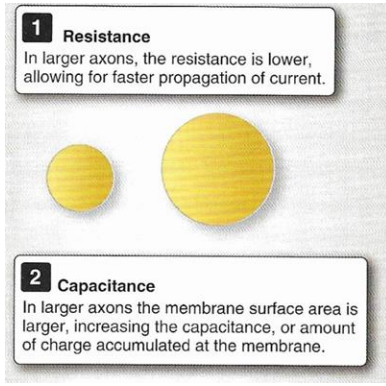


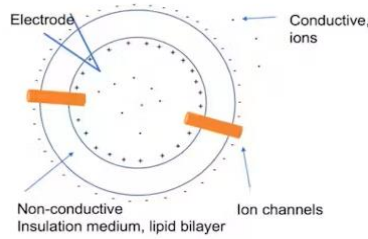
What factors determine the conduction velocity of axons?

The spread of current along an axon from an active region to an inactive region depends on three factors:

- The resistance of the axon to the flow of electrical current along its length (its internal electrical resistance)\: less internal resistance [Ri] for larger diameter
- The electrical resistance of the axon membrane [Rm] NB: effect of myelin increases Rm
- The electrical capacitance of the axon membrane (capacitance is measured per unit length of the axon); α to diameter of axon i.e. the larger the membrane area the more charge it can hold.







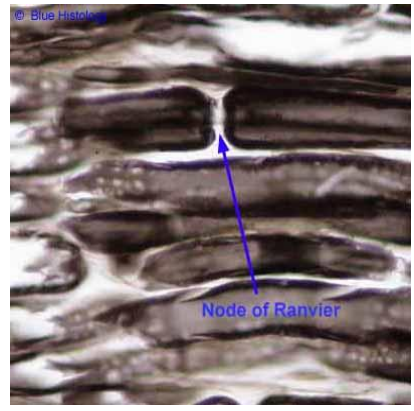
Capacitance:
Determines how quickly the membrane potential can respond to a change in current
Lipid membrane = insulator, the IC and EC fluids are the conductors
When there is a voltage difference across an insulator, charge will build up at the interface (current cannot flow across the insulator)
The larger the cell the larger the capacitance will be……
But capacitance is directly proportional to the distance between the conducting media- thus *myelinated* axons have smaller capacitance
Axonal Internal Resistance is inversely proportional to d^2
Capacitance increases in direct proportion to 'd'

$$Velocity \quad \alpha \quad \frac{1}{C_m}\sqrt{\frac{d}{4R_mR_i}}$$

43

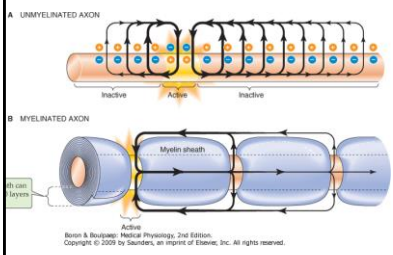
TWO main mechanisms have emerged to increase speed of conduction

1. increase the **diameter** of the nerve (CV is proportional to the sq rt of the axon diameter. 1=1; 2=1.4m/s; 4= 2m/s etc)
2. **myelination** (insulation; nodes of Ranvier- i. 'saltatory' conduction; ii. increased insulation decreases capacitance thus resistance;) CV is directly proportional to the myelin thickness.



44

Action potential: Mechanisms of conduction



Unmyelinated: Continuous conduction (current flows from A (active region) to B. Like a fuse!

'Saltatory' conduction: myelinated nerves- current can only cross the membrane at the nodes

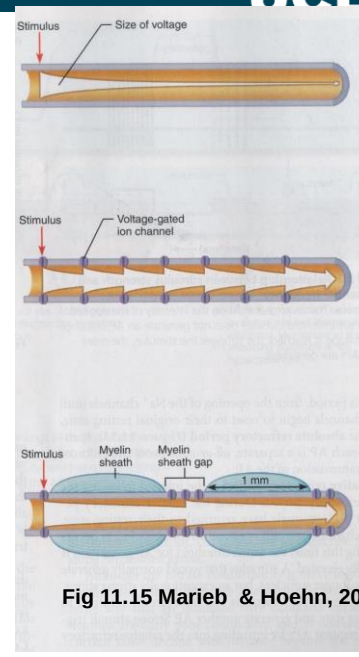
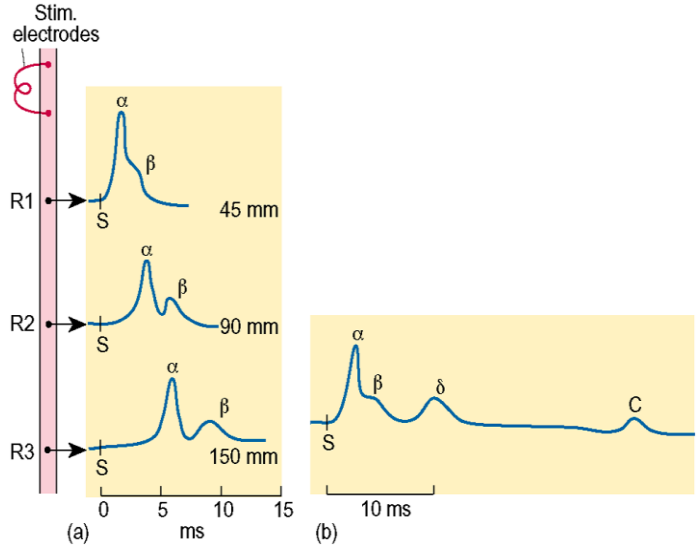


Fig 11.15 Marieb & Hoehn, 2013

Action potential: Mechanisms of conduction

Compound action potentials show that some axons conduct faster than others



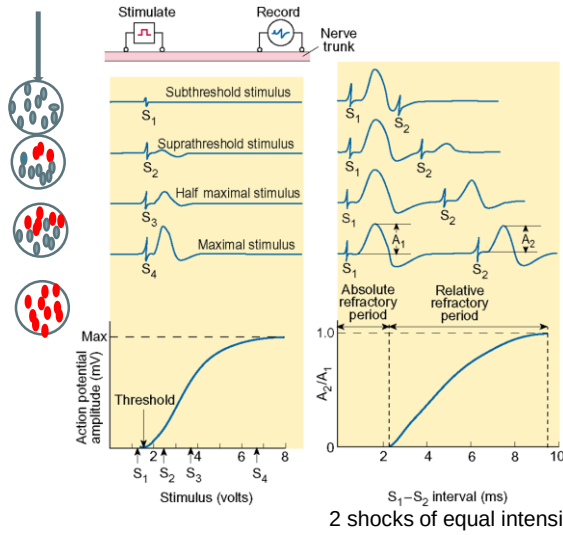
'A' fibres conduct at 30-70m/s depending on type (sensory afferents are fastest).

'C' fibres are the slowest: 0.5 - 2.0m/s. These are unmyelinated

46

Action potential: Mechanisms of conduction

Action potentials can be recorded ***from intact nerves***: **compound action potentials (CAP) when record from several axons**



2 shocks of equal intensity

The CAP represents the result of stimulation of several axons, hence its change in amplitude with increasing stimulus strength.

There is also a refractory period for whole nerve stimulation.

47

Fast and slow spiking neurons……

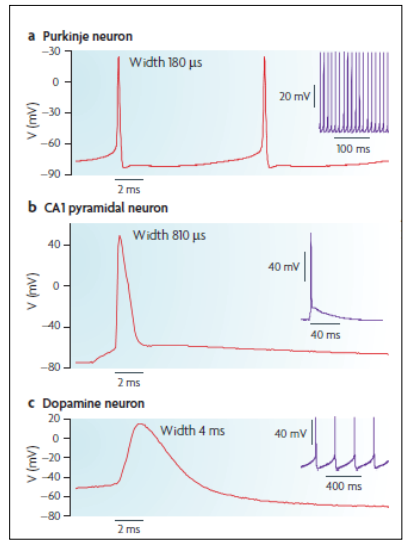
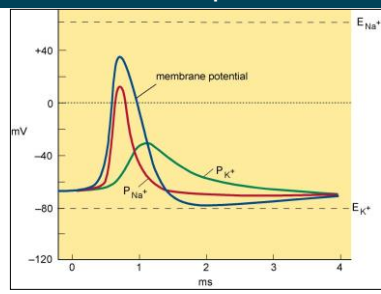


Fig.1 Bean, 2007

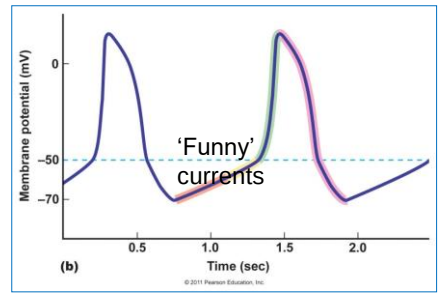
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48

Neuronal action potential

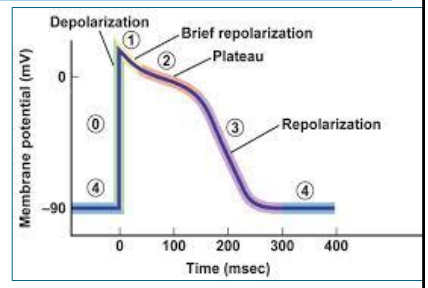


Cardiac action potentials



4 key points:
Threshold, time course and duration of the AP depend on:

1. The gating and permeability of specific types of ion channels- these depend on Vm and time.
2. The intracellular and extracellular concentrations of ions that pass through those channels: Na+, K+, Ca+ Cl-
3. Membrane properties such as capacitance, resistance and geometry of the cell
4. The shape of the AP reflects the specialized function of that cell



49

Summary 2:

- Neurons are the principal functional units of the nervous system
- Neurons send signals to other neurons and target cells by means of action potentials, propagated along their axons, which are typically unidirectional and their ***frequency codes*** information in the NS.
- Action potentials are generated when a neuron is activated by a stimulus of a certain minimum strength known as the threshold.
- Each type of action potential e.g. neuronal, cardiac myocyte etc has approximately the same magnitude and duration. This is known as the "all or none" law.
- The action potential is caused by a large, short-lived increase in the permeability of the membrane to ***sodium ions***. Self-propagating.
- After the passage of one action potential, the axon cannot propagate another until sufficient sodium channels have returned to their resting state. This period of inexcitability is known as the absolute refractory period and lasts for 1-2 m (in neurons). After the ***absolute*** refractory period there is a time of reduced excitability called the ***relative*** refractory period.
- Myelinated nerves conduct more quickly via 'saltatory' conduction

50

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Further reading:

Pract Neurol 2007; 7: 192–197
The action potential
Mark W Barnett, Philip M Larkman

NATURE REVIEWS | NEUROSCIENCE VOLUME 8 | JUNE 2007 |
The action potential in mammalian central neurons
Bruce P. Bean

51